

Saurabh Jalendra

Research Engineer working at the intersection of reinforcement learning, world models, LLM systems, and agentic systems.

M.Tech AI/ML; 2 years shipping production AI products.

saurabhjalendra@gmail.com | [+91-9783982649](tel:+91-9783982649) | Jaipur, India | Open to relocate | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Experience

Founding Research Engineer (Co-Founder) – SKY Advanced Research LLP, Jaipur Oct 2024 – Present

Klares, Hong Kong | Investor Relations Platform and Document Intelligence

- Engineered DDQ Autofill closed-loop: vision-LLM extracts questions from complex Due Diligence Questionnaires (multi-section, nested; DOCX/XLSX/PDF), generates review UI matching the original layout, autogenerates answers via RAG pipeline (hybrid dense + cross-encoder reranking) over a knowledge base ingested from fund/investor/broker documents with LLM-as-judge quality scoring, exports the filled Questionnaire in original format, cutting DDQ turnaround from a month to minutes.
- Shipped investor-relations workflow: Microsoft Graph Outlook email ingestion (investors/fund managers/brokers), document classification + chronological version control, date/period extraction for renewal/expiry reminders, auto-requirement extraction with custom document-pack sharing, cutting requirement-settlement turnaround from weeks to minutes.
- Deployed on Google Cloud with Arize Phoenix observability, GCS object storage, RBAC, and audit logging.

International Citizen, Hong Kong | Cross-border portfolio and wealth management platform for NRI/expat investors with multilingual statement support (Hindi/Chinese/English) (team-built; led extraction + conversational AI)

- Vision-LLM multi-format document extraction + ingestion pipeline (PDF, CSV, XLSX, images): auto-classifies statement type (bank/fund/investment), extracts transactions + holdings + metadata into a structured vector database, auto-deduplicates against existing accounts and updates transactions incrementally.
- SSE-streaming conversational AI with RAG over the vector store: portfolio queries, real-market questions, and on-demand report generation; real-time portfolio dashboard with multi-currency aggregation across global exchanges segregated by institution/holder, live pricing + profit/gain tracking.

KAT GCS, Jaipur | Drone Ground Control System (team-built; co-architected real-time stack)

- Electron desktop GCS for drone fleet operations: real-time MAVLink telemetry, mission planning with obstacle avoidance, geofencing, multi-vehicle support, and live AI video analysis.

Operations and Business Development – S K Jalendra Marketing Services Pvt Ltd, Jaipur Dec 2017 – Mar 2024

- Channel partner to Banks/NBFCs; led operations, team management, and client relationships.

Research Intern – ISRO (National Remote Sensing Centre), Jodhpur Feb 2017 – Jun 2017

- Built interactive Virtual Reality heritage simulation with remote-controller navigation and gaze-triggered explanations of artifacts presented in the scene (Unity Game Engine, Oculus Rift, HTC Vive); 3D artifact models generated via 360° photogrammetry of museum statues (Meshroom).

Research Projects

Simulating Anything Feb 2026 – Present

- Domain-agnostic scientific discovery pipeline: natural language query → simulation → RSSM world model (JAX/Equinox) → symbolic regression → human-interpretable equations.
- Implemented 187 simulation domains across physics/biology/chemistry; PySR/SINDy recovered physical laws with $R^2 \geq 0.999$ on 11/14 core domains (mean $R^2 = 0.970$); 570 cross-domain isomorphisms detected across 125 equivalence classes.

Quantum-Enhanced Simulation Learning for RL Nov 2025 – Feb 2026

- M.Tech dissertation (defended): implemented 4 quantum-inspired methods (tunneling optimizer, superposition buffer, entanglement layer, interference ensemble) on DreamerV3-style world models across 8 environments (CartPole, Pendulum, Walker-walk, Cheetah-run, Reacher-easy/hard, Pong, Breakout).
- Interference Ensemble: 36–47% reduction in world model prediction MSE on DMControl (Walker, Cheetah, Reacher-easy/hard; $p < 0.008$, Mann-Whitney U Bonferroni-corrected); 200 experiments across 5 methods × 5 seeds × 8 environments.

Education

Birla Institute of Technology and Science, Pilani – M.Tech in AI/ML Jun 2024 – May 2026

- Dissertation:** Quantum-Enhanced Simulation Learning for RL

NMIMS, Mumbai – MBA in Banking and Finance Jan 2022 – Jan 2024

Government Engineering College, Bikaner – B.Tech in Computer Science Engineering May 2013 – May 2020

Technical Skills

- Languages:** Python, TypeScript, SQL
- ML/DL:** PyTorch, JAX/Equinox/Optax, scikit-learn, Gymnasium, DMControl, PySR/PySINDy
- LLM/Retrieval:** LlamaIndex, Hugging Face, RAG pipelines, vector databases (pgvector)
- Web/Backend:** FastAPI, React, Node.js, PostgreSQL/SQLite
- Data/Viz:** Pandas, NumPy, Matplotlib
- Cloud/DevOps:** AWS, Google Cloud Platform (GCS, Cloud Run), Docker, GitHub Actions, CI/CD
- AI Dev Tools:** Claude Code (agent orchestration, skill authoring, MCP servers, slash commands)